

$$P = \begin{bmatrix} 0 & 0.25 \\ 1 & 0.75 \end{bmatrix}$$

$$X_0 = \begin{bmatrix} 200 \\ 200 \end{bmatrix}$$

$$a) X_1 = P X_0: \begin{bmatrix} 0 & 0.25 \\ 1 & 0.75 \end{bmatrix} \cdot \begin{bmatrix} 200 \\ 200 \end{bmatrix} = \begin{bmatrix} 50 \\ 350 \end{bmatrix}$$

So, 50 students will watch 1 hour or more tomorrow.

$$b) X_2 = P^2 X_0: \begin{bmatrix} 0 & 0.25 \\ 1 & 0.75 \end{bmatrix}^2 = \begin{bmatrix} 0.25 & 0.1875 \\ 0.75 & 0.8125 \end{bmatrix} \cdot \begin{bmatrix} 200 \\ 200 \end{bmatrix} = \begin{bmatrix} 87.5 \\ 312.5 \end{bmatrix}$$

So, 88 students will watch 1 hour or more in 2 days.

$$c) X_{40} = P^{40} X_0: \begin{bmatrix} 0 & 0.25 \\ 1 & 0.75 \end{bmatrix}^{40} = \begin{bmatrix} 0.2 & 0.2 \\ 0.8 & 0.8 \end{bmatrix} \cdot \begin{bmatrix} 200 \\ 200 \end{bmatrix} = \begin{bmatrix} 80 \\ 320 \end{bmatrix}$$

So, 80 students will watch 1 hour or more in 40 days.